

SpaceX Falcon 9 First Stage Landing Prediction

Project Overview

This project predicts whether the **Falcon 9 first stage booster will successfully land** after a launch. SpaceX advertises Falcon 9 rocket launches at approximately \$62 million, compared to competitors who charge upward of \$165 million per launch — a large part of the savings comes from reusing the first stage. By predicting landing success, we can estimate the cost of a launch and gain competitive intelligence.

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Dataset

The project uses three datasets sourced from IBM Skills Network:

Dataset	Description
dataset_part_1.csv	Raw launch data — orbits, outcomes, launch sites
dataset_part_2.csv	Engineered features including payload mass, grid fins, reuse counts
dataset_part_3.csv	One-hot encoded feature matrix for ML models
spacex_launch_geo.csv	Geospatial data for launch site mapping
Spacex.csv	Full mission data loaded into SQL

The **target variable** is **Class**:

- ① → Successful landing
- ① → Failed landing

Project Pipeline

Raw Data → EDA & SQL Analysis → Feature Engineering → Visualization → ML Modeling
→ Evaluation

Technologies Used

- Python 3
- Pandas — data manipulation
- NumPy — numerical operations
- Seaborn / Matplotlib — static visualizations
- Plotly Express — interactive charts
- Folium — geospatial map visualizations
- SQLite + ipython-sql — SQL-based data exploration
- Scikit-learn — machine learning models and evaluation

Modules

1. Data Collection & Wrangling

File: `Falcon9__final__1_.ipynb` (Part 1)

- Loaded launch data from a remote CSV
- Inspected missing values (calculated as % of total)
- Examined data types with `df.dtypes`
- Counted value distributions for `LaunchSite`, `Orbit`, and `Outcome`
- Created the binary `Class` column by classifying outcomes as success (1) or failure (0)

```
python

bad_outcome = set(landing_outcomes.keys()[[1, 3, 5, 6, 7]])
landing_class = [0 if outcome in bad_outcome else 1 for outcome in df['Outcome']]
df['Class'] = landing_class
```

2. SQL Analysis

File: `Falcon9__final__1_.ipynb` (Part 2)

- Loaded SpaceX data into an **SQLite database** (`my_data1.db`)
- Performed SQL queries to extract insights:

Query	Purpose
<code>DISTINCT Launch_Site</code>	List unique launch sites
<code>SUM(Payload_Mass__kg_) for NASA (CRS)</code>	Total payload for a customer
<code>AVG(Payload_Mass__kg_) for F9 v1.1</code>	Average payload by booster version
<code>MIN(Date) for ground pad success</code>	First successful ground pad landing
Booster versions with drone ship success + high payload	Filter by multiple conditions
Landing outcomes between 2010–2017	Count outcomes over a date range

3. Exploratory Data Analysis (EDA)

File: `Falcon9__final__1_.ipynb` (Part 3)

Visualizations produced using **Seaborn**:

- **Flight Number vs Payload Mass** — colored by landing class
- **Flight Number vs Launch Site** — which sites improve over time
- **Payload Mass vs Launch Site** — mass ranges per site
- **Orbit vs Success Rate** — bar chart of mean success per orbit type
- **Flight Number vs Orbit** — success distribution across orbits
- **Payload Mass vs Orbit** — mass impact on orbit success
- **Yearly Success Rate** — line chart of landing success trend over years

Key finding: Success rate improves significantly over the years as SpaceX iterates on technology.

4. Interactive Visual Analytics

File: `Falcon9__final__1_.ipynb` (Part 4)

Folium Map

- Plotted all Falcon 9 launch sites on an interactive map
- **Green star** = Successful landing
- **Red X** = Failed landing
- Sites covered: `CCAFS LC-40`, `CCAFS SLC-40`, `KSC LC-39A`, `VAFB SLC-4E`

Plotly Charts

- **Bar chart** — Success vs Failure count by Launch Site
 - **Pie chart** — Overall launch success rate across all missions
-

5. Machine Learning Prediction

File: `Falcon9__final__1_.ipynb` (Part 5)

Feature Engineering

- Selected features: `FlightNumber`, `PayloadMass`, `Orbit`, `LaunchSite`, `Flights`, `GridFins`, `Reused`, `Legs`, `LandingPad`, `Block`, `ReusedCount`, `Serial`
- Applied **one-hot encoding** on categorical columns
- Scaled features with `StandardScaler`
- Split data: **80% train / 20% test** (`random_state=2`)

Models Trained (with GridSearchCV + 10-fold CV)

Model	Hyperparameters Tuned
Logistic Regression	<code>C</code> , <code>penalty</code> , <code>solver</code>
Support Vector Machine (SVM)	<code>kernel</code> , <code>C</code> , <code>gamma</code>
Decision Tree	<code>max_depth</code> , <code>criterion</code>
K-Nearest Neighbors (KNN)	<code>n_neighbors</code> , <code>algorithm</code> , <code>p</code>

Model Results

All models were evaluated on the **held-out test set** using accuracy score and a confusion matrix.

```

Model                               : Test Accuracy
-----:-----
Logistic Regression                  : (see notebook output)
SVM                                  : (see notebook output)
Decision Tree                        : (see notebook output)
KNN                                  : (see notebook output)

```

Run the notebook to see exact accuracy scores and confusion matrix visualizations.

The confusion matrix labels:

- `land` — Booster successfully landed
- `did not land` — Booster failed to land

How to Run

1. **Clone or download** this notebook.
2. **Install dependencies:**

```
bash  
pip install pandas numpy seaborn matplotlib scikit-learn folium plotly sqlalchemy ip
```

3. **Open the notebook:**

```
bash  
jupyter notebook Falcon9__final__1_.ipynb
```

4. **Run all cells** in order. The datasets are fetched automatically from IBM's public cloud storage — no manual download required.

Project Structure

```
Falcon9-Landing-Prediction/  
|  
├─ Falcon9__final__1_.ipynb    # Main project notebook  
├─ dataset_part_3.csv          # Generated during EDA (one-hot encoded features)  
├─ my_data1.db                 # SQLite database (auto-created)  
└─ README.md                   # This file
```

Key Insights

- SpaceX landing success rate has **increased dramatically** over the years.
- **KSC LC-39A** tends to have a higher success rate compared to other launch sites.
- **Higher flight numbers** generally correlate with more successful landings (experience effect).
- **Orbit type** strongly influences success — LEO and ISS missions tend to perform better.
- Machine learning models achieve strong accuracy in predicting landing outcomes using structured launch data.

